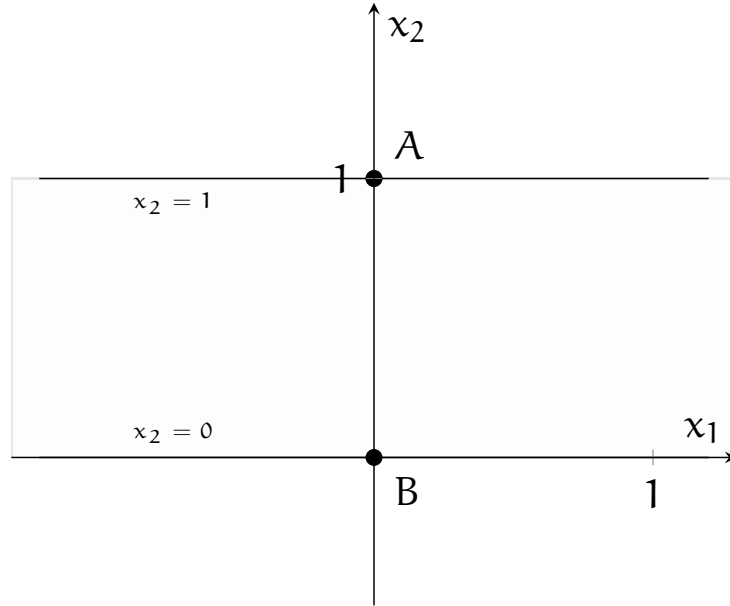


The constraint polyhedron is represented in the figure below:



Note that this polyhedron in $\mathbb{R}^2$ has no vertex. In order to write the problem in standard form, we need to replace x_1 by the difference of two non negative variables

$$x_1 = x_1^+ - x_1^-,$$

and we need to introduce a slack variable for the constraint $x_2 \leq 1$. We obtain

$$\min_{x \in \mathbb{R}^4} x_1^+ - x_1^- + x_2$$

subject to

$$\begin{aligned} x_2 + x_3 &= 1, \\ x_1^+, x_1^-, x_2, x_3 &\geq 0. \end{aligned}$$

In matrix form, we have

$$\min_{x \in \mathbb{R}^4} c^T x$$

subject to

$$\begin{aligned} Ax &= b, \\ x &\geq 0, \end{aligned}$$

where

$$A = (0 \ 0 \ 1 \ 1),$$

$$b = 1,$$

$$c = \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix},$$

$$x = \begin{pmatrix} x_1^+ \\ x_1^- \\ x_2 \\ x_3 \end{pmatrix}.$$

This is a problem with $n = 4$ variables and $m = 1$ constraints. Therefore, there are four potential bases.

1. x_1^+ in the basis. In this case $B = 0$. It is not invertible. Therefore, it is not a valid basis.
2. x_1^- in the basis. In this case $B = 0$. It is not invertible. Therefore, it is not a valid basis.
3. x_2 is in the basis. In this case, $B = 1$ and

$$x_2 = B^{-1}b = 1.$$

It is feasible. The vertex

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix},$$

corresponds to point A on the figure. Note that it is a vertex of the polyhedron in $\mathbb{R}^4$, even it does not correspond to a vertex of the original polyhedron in $\mathbb{R}^2$.

4. x_3 is in the basis. In this case, $B = 1$ and

$$x_3 = B^{-1}b = 1.$$

It is feasible. The vertex

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

corresponds to point B on the figure. Again, it is a vertex of the polyhedron in $\mathbb{R}^4$, even it does not correspond to a vertex of the original polyhedron in $\mathbb{R}^2$.